

Food Hub Data Analysis

PGP-AIML-BA-UTA-JAN'26-C Program Project 1

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Executive Summary

Introduction and Context

FoodHub operates a commission-based food aggregation platform connecting customers with restaurants across New York. This analysis evaluates 1,898 customer orders across 178 restaurants and 1,200 customers. The purpose of this analysis is to provide insights to assist the company attain its stated goals of improving customer satisfaction and increasing revenue.

Summary of Findings

This analysis finds that the company's strongest growth opportunity is by increasing engagement and retention of customers, encouraging more high value orders, and improving operational efficiency to reduce the total fulfillment time of orders.

Strategic Recommendations

- Implement Loyalty-Based Incentives to increase repeat frequency and deepen platform-centered retention.
- Transform Ratings into Engagement Tools by linking ratings to rewards or discounts.
- Optimize Weekday Delivery Logistics through driver allocation and routing improvements.
- Promote High-Rated, High-Retention Restaurants in marketing campaigns.
- Restructure the tiered commission system to:
 - Improve margin efficiency on lower-value orders.
 - Assess the value of implementing a minimum order requirement.
- Consider incentives to encourage higher order values.

Business Problem Overview and Solution Approach

Analysis of Customer Characteristics

Findings

- American cuisine is the most popular overall and dominates weekend demand.
- Revenue is concentrated among a small subset of high-performing restaurants.
- Only 34.7% of customers are repeat users, yet they are responsible for 21.9 % of the total orders and 58.7% of total revenue.
- Repeat customers do not spend more per order; revenue concentration is driven by frequent orders and not the cost of the order
- The Top 5 customers placed between 7–13 orders each

Business Insights

- Customer Retention is a major revenue driver. Since repeat customers do not spend more per order, increasing order frequency is the most effective means to increase revenue.
- Repeat customers have a stronger preference for Japanese cuisine compared to single-order customers.
- American, Italian, Chinese, Mexican cuisines are more popular among single-order customers

Strategic Opportunities

- Identify the restaurants can improve both revenue and customer retention. Finding and strengthening partnerships with these restaurants is strategically valuable.

Business Problem Overview and Solution Approach

Analysis of Business Operations

Findings

- Mean delivery time: 24.16 minutes.
- Mean weekday delivery: ~28 minutes.
- Mean weekend delivery: ~22 minutes.
- 10.54% of orders exceed 60 minutes fulfillment time.
- Orders that cost less than \$5 are delivered free.

Business Insights

- Delivery times are slower on weekdays, indicating potential logistics inefficiencies.
- Service time does not significantly impact ratings or repeat customer behavior.
- Ratings participation is moderate (61% rated, 39% unrated) but appears uncorrelated with the cost of the order or total delivery time.

Business Risks

- Weekday inefficiency suggests potential driver allocation or routing challenges. Orders that take more than 60 minutes to fulfill negatively impact customer satisfaction.

Strategic Opportunity

- Consider revising the current tiered commission system. The current system incentivizes participation of low order cost restaurants.
- Consider implementing a minimum order value that covers the cost of delivery.

Business Problem Overview and Solution Approach

Analysis of Rating System

Findings

- 736 orders were not rated (38.8%).
- Average rating among rated orders is high (~4.3+).

Business Insights

- Ratings do not significantly correlate with repeated customer behavior, service times, or order costs (revenue).
- Metrics other than Customer ratings should be considered for promotion eligibility criteria.
- Ratings participation is moderate (61% rated, 39% unrated) but appears uncorrelated with the cost of the order or total delivery time.

Strategic Opportunity

Ratings currently function as passive feedback tools. Integrating ratings into a loyalty incentive system could increase engagement and repeat behavior.

Data Overview

Dataset Characteristics

- The original dataset (df) is comprised of 1,898 orders, 9 variables, and no missing values.
- The original dataset (df) is comprised of 5 numerical variables and 4 categorical variables.
 - The values of columns “customer_id” and “order_id” are categorical labels rather than numerical variables. Although it is not required for this project, the values in these columns should be changed from int to string.
- The “Rating” variable is categorical rather than numerical since the value “Not Given” is a string.
 - Although it is not required for this project, the string value should be removed and replaced with NaN to allow numerical analysis of rating data.

Summary of categorical variables

- Unique customers IDs = 1200
- Unique restaurant names = 178
- Unique cuisine types = 14
- Unique days of the week = 2

Specified Tasks

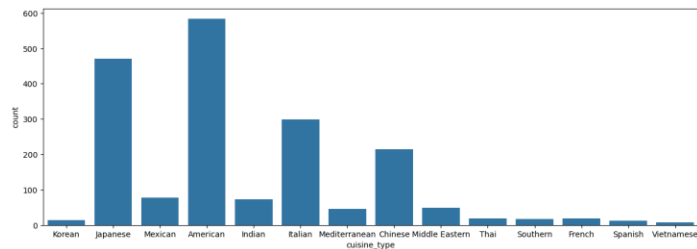
The minimum, maximum, and average food preparation time is 20.0 minutes, 35.0 minutes, 27.4 minutes, respectively.

The number of orders not rated is 736 or 39% of all orders.

Univariate Analysis

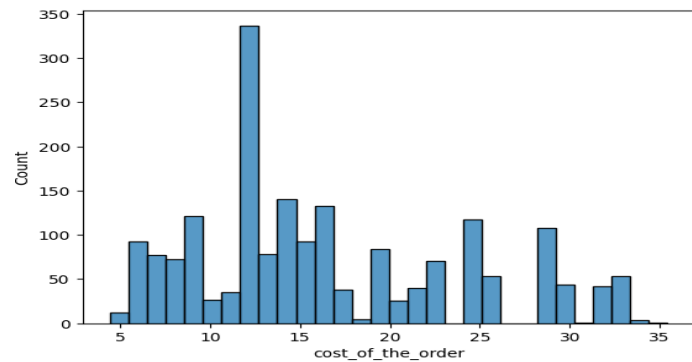
Distribution of Orders by Cuisine type

- American, Japanese, Italian, and Chinese are the most preferred cuisine type



Distribution of orders by cost

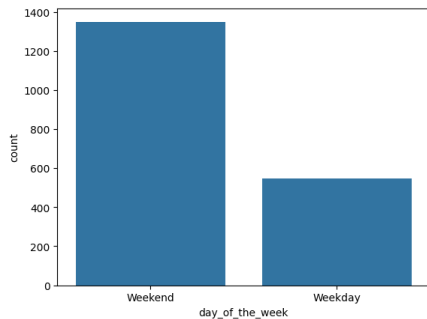
- The distribution shows a clustering of the costs of the order that warrant further investigation.



Univariate Analysis

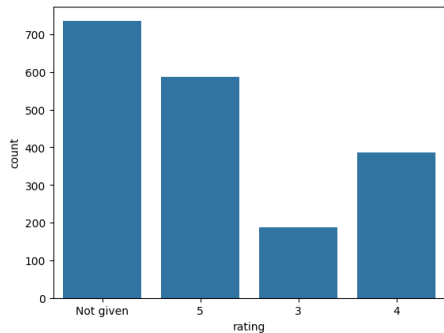
Distribution of orders by Day of the Week

- The number of orders placed on weekends is significantly higher than the number of orders placed on weekdays.



Distribution of orders by Rating

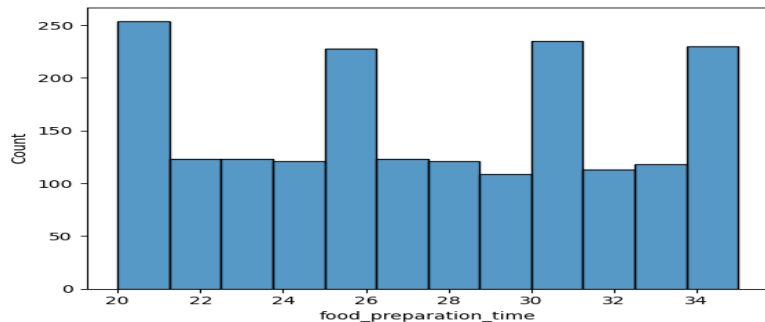
- There is a significant number of orders that are not rated.



Univariate Analysis

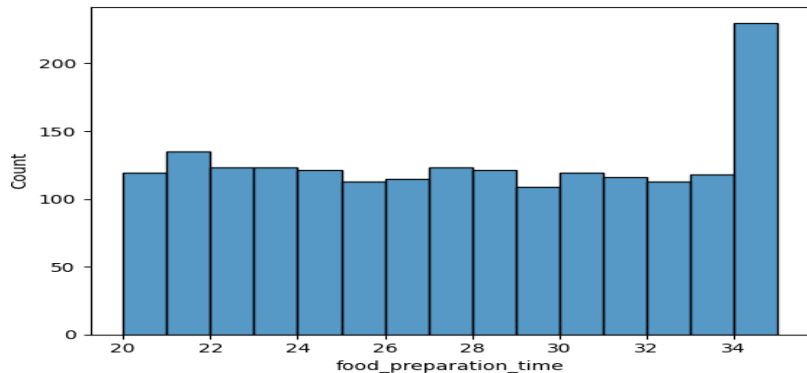
Distribution of orders by Food Preparation Time

There are four spikes when the histogram is plotted using default setting



However, a different pattern emerges when the bin width is reduced to 1-minute intervals.

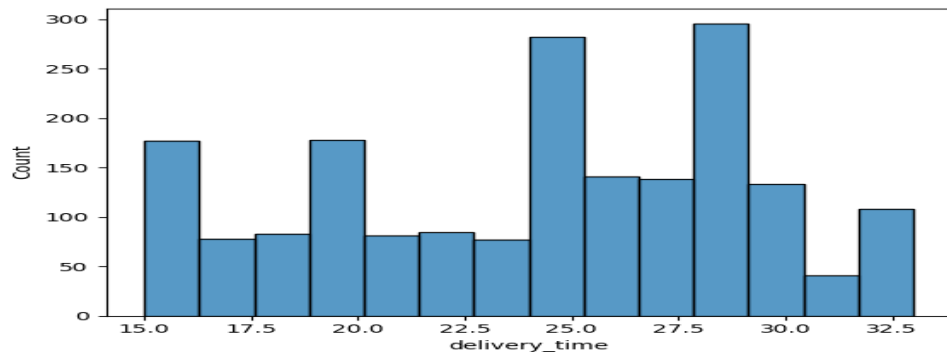
- There is a significant spike of orders at the maximum reported food preparation time.



Univariate Analysis

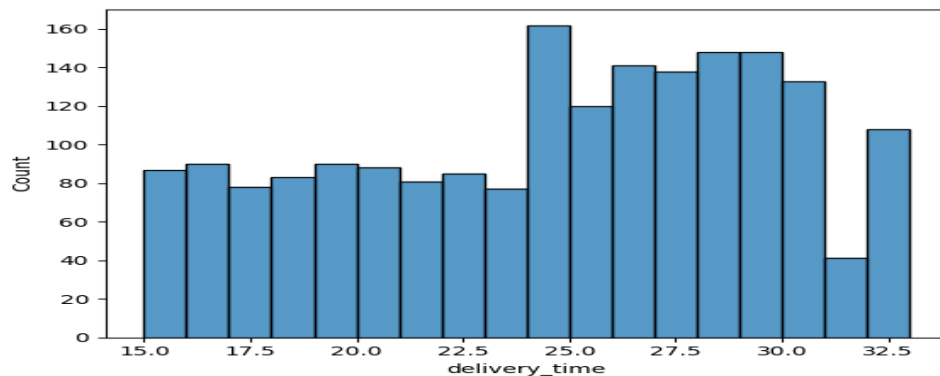
Distribution of orders by Delivery Time

There are four spikes when the histogram is plotted using default setting



However, a different pattern emerges when the bin width is reduced to 1-minute intervals.

- There is a pronounced clustering between 24 and 31 minutes.



Univariate Analysis

Questions:

Which are the top 5 restaurants in terms of the number of orders received?

restaurant_name	
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

Which is the most popular cuisine on weekends?

cuisine_type	
American	415
Japanese	335
Italian	207
Chinese	163
Mexican	53
Indian	49

Univariate Analysis

Questions:

What percentage of the orders cost more than 20 dollars?

- The number of total orders that cost above 20 dollars is: 555
- Percentage of orders above 20 dollars: 29.24 %

What is the mean order delivery time?

- The mean delivery time for this dataset is 24.16 minutes

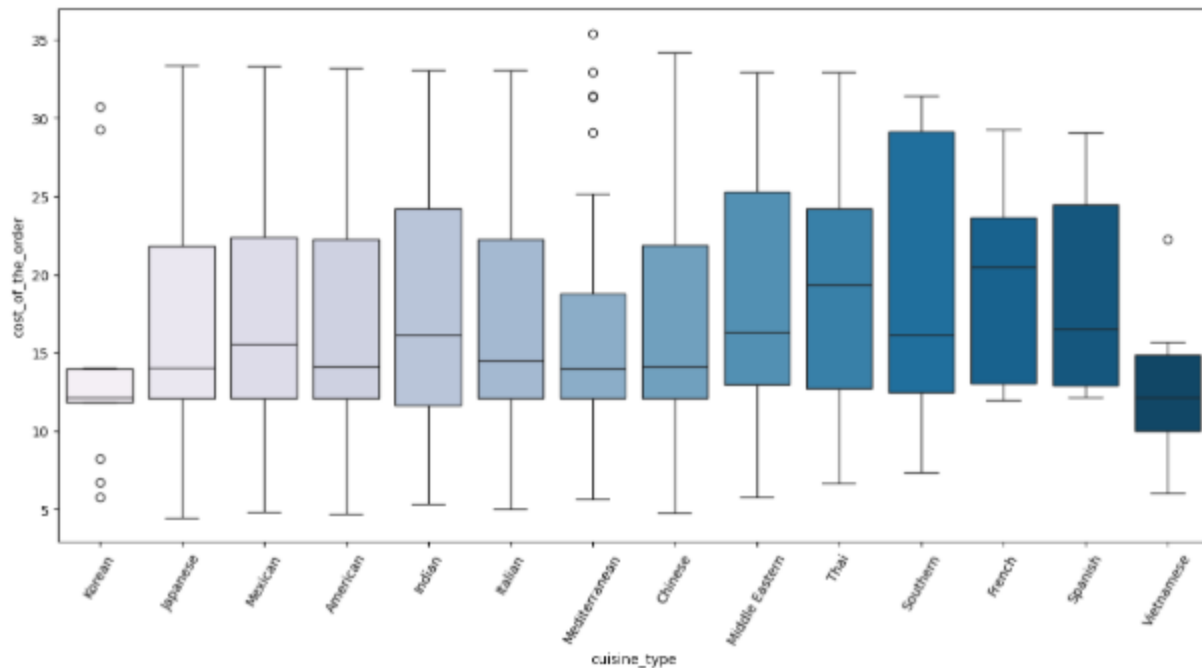
The company has decided to give 20% discount vouchers to the top 5 most frequent customers. Find the IDs of these customers and the number of orders they placed.

customer_id	
52832	13
47440	10
83287	9
250494	8
259341	7

Multivariate Analysis

Cuisine vs Cost of the order

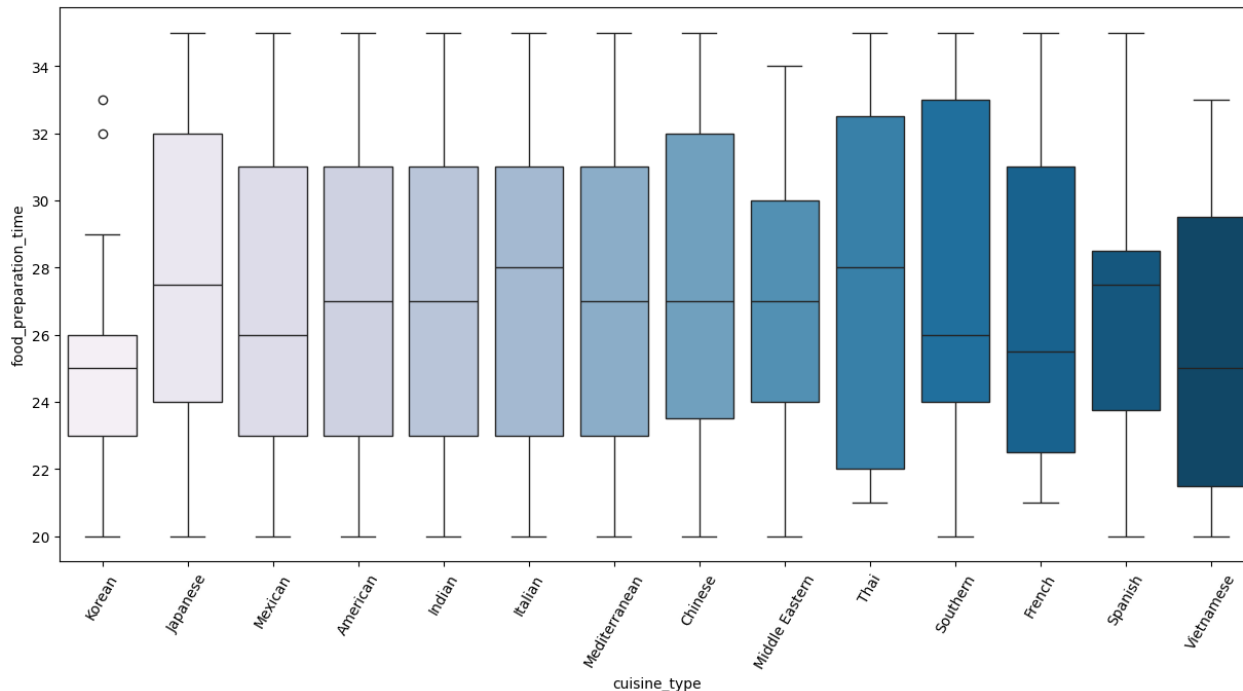
The data show that for most cuisines there is a left-skewed distribution regarding the cost of the order. This observation warrants further analysis. Many small orders and fewer large orders.



Multivariate Analysis

Cuisine vs Food Preparation time

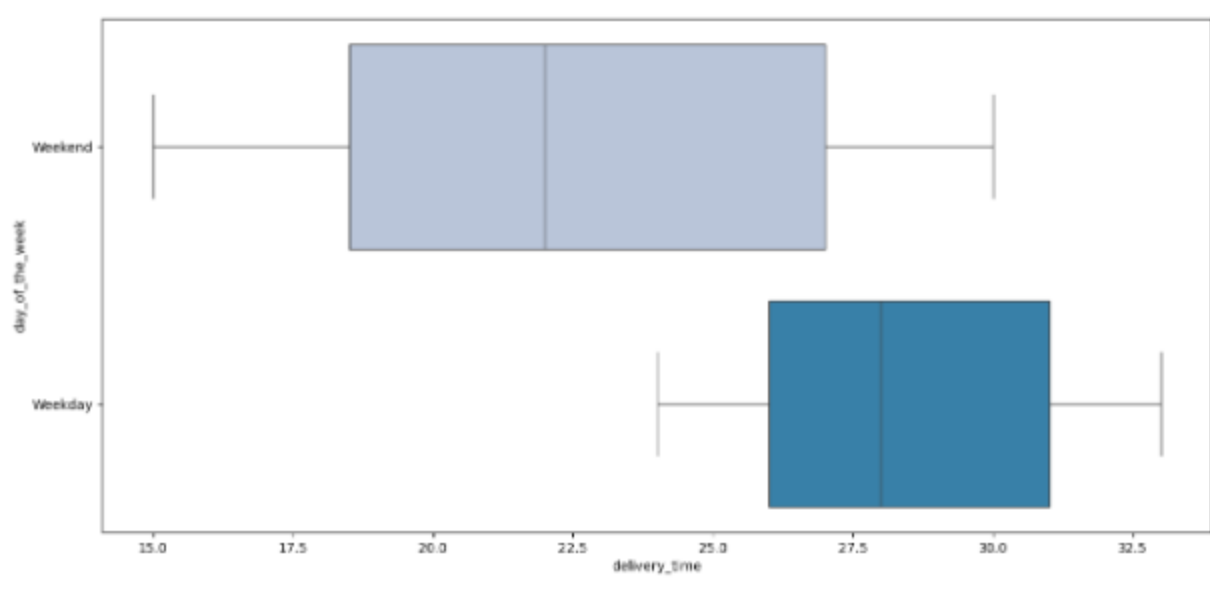
Korean Cuisine has shortest prep time. Thai cuisine has high variability



Multivariate Analysis

Day Of The Week And Delivery Time

Delivery times are much higher on weekdays than on weekends



Multivariate Analysis

Use the provided code to make observations about the revenue generated by the restaurants.

- The provided code only describes the sum of the cost of orders. Revenue uses a tiered commission scheme that applies a variable percentage based on the order size. Revenue by restaurant cannot be determined by this code.
- The sum cost of orders is dominated by a small number of restaurants.

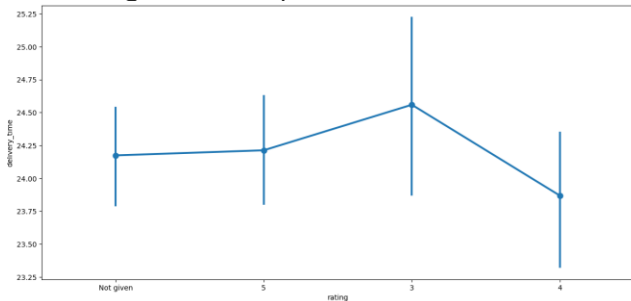
restaurant_name	cost_of_the_order
Shake Shack	3579.53
The Meatball Shop	2145.21
Blue Ribbon Sushi	1903.95
Blue Ribbon Fried Chicken	1662.29
Parm	1112.76
RedFarm Broadway	965.13
RedFarm Hudson	921.21
TAO	834.50
Han Dynasty	755.29
Blue Ribbon Sushi Bar & Grill	666.62
Rubirosa	660.45
Sushi of Gari 46	640.87
Nobu Next Door	623.67
Five Guys Burgers and Fries	506.47
Momoya	492.13
Chipotle Mexican Grill \$1.99 Delivery	491.69
Blue Ribbon Sushi Izakaya	485.09
Tamarind TriBeCa	426.71
Jack's Wife Freda	416.75
Bareburger	399.74

Multivariate Analysis

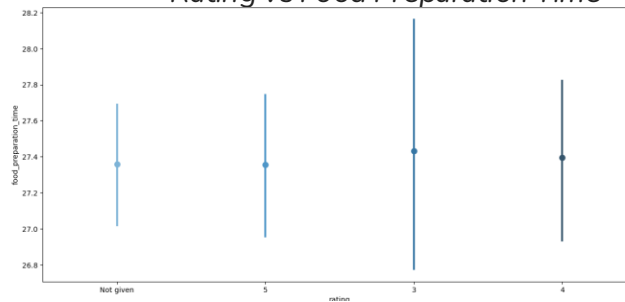
Correlation Analysis of Ratings

- There is no strong correlation between Ratings and other critical business metrics

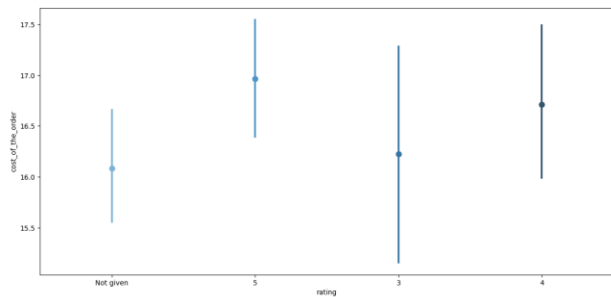
Rating vs Delivery Time



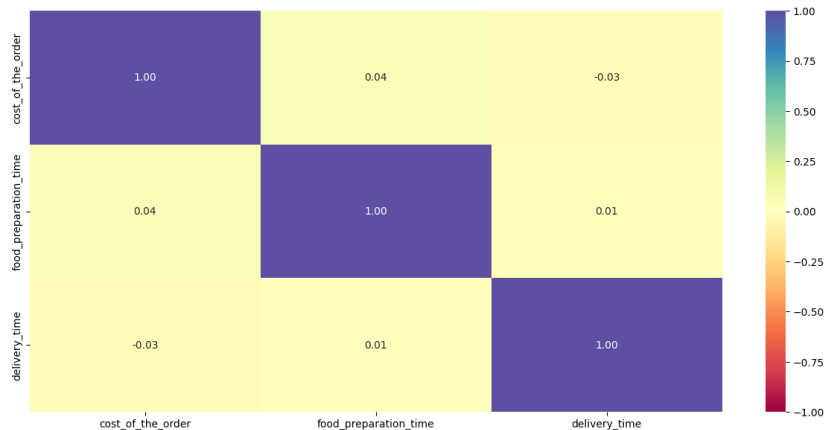
Rating vs Food Preparation Time



Rating vs Cost of Order



Correlation Matrix



Multivariate Analysis

To qualify for the promotional offer, restaurants must have a rating count of more than 50 and the average rating should be greater than 4.

	restaurant_name	rating
0	Shake Shack	133
1	The Meatball Shop	84
2	Blue Ribbon Sushi	73
3	Blue Ribbon Fried Chicken	64

	restaurant_name	rating
0	The Meatball Shop	4.511905
1	Blue Ribbon Fried Chicken	4.328125
2	Shake Shack	4.278195
3	Blue Ribbon Sushi	4.219178

The qualifying restaurants are:

- Shake Shack
- The Meatball Shop
- Blue Ribbon Sushi
- Blue Ribbon Fried Chicken

Multivariate Analysis

The company charges the restaurant 25% on the orders having cost greater than 20 dollars and 15% on the orders having cost greater than 5 dollars. Determine the net revenue generated by the company across all orders.

- The net revenue is around 6166.3 dollars

The company wants to analyze the total time required to deliver the food. What percentage of orders take more than 60 minutes to get delivered from the time the order is placed?

- The number of orders with a total time of more than 60 minutes is 200 orders.
- Percentage of orders taking more than 60 minutes: 10.54%

The company wants to analyze the delivery time of the orders on weekdays and weekends. How does the mean delivery time vary during weekdays and weekends?

- The mean delivery time on weekdays is around 28 minutes
- The mean delivery time on weekends is around 22 minutes



Happy Learning !

